# CONTROL ENGINEERING

1. a) The matrices $A$, $B$, $C$, and $D$ are given by

$$A = \begin{bmatrix} 1-\alpha_1-\delta_1 & 0 & 0 \\ \alpha_1 & 1-\alpha_2-\delta_2 & \beta_3 \\ 0 & \alpha_2 & 1-\beta_3-\gamma_3 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix},$$

$C = \begin{bmatrix} 0 & 0 & \gamma_3 \end{bmatrix}$, and $D = 0$.

*Incorrect dimensions of the matrices and/or incorrect entries.*

[ 4 marks ]

b) The reachability matrix of the system is

$$\mathscr{R} = \begin{bmatrix} 1 & 1-\alpha_1-\delta_1 & (1-\alpha_1-\delta_1)^2 \\ 0 & \alpha_1 & \alpha_1(1-\alpha_1-\delta_1)+\alpha_1(1-\alpha_2-\delta_2) \\ 0 & 0 & \alpha_1\alpha_2 \end{bmatrix}.$$

The system is therefore reachable provided $\alpha_1\alpha_2 \neq 0$.

*Incorrect entries, incorrect computation of the determinant/rank of $\mathscr{R}$, and/or incorrect conclusions.*

[ 4 marks ]

c) If $\beta_3 = 0$ then the eigenvalues of the $A$ matrix are $1-\alpha_1-\delta_1$, $1-\alpha_2-\delta_2$, and $1-\gamma_3$. Note that, as stated in the exam paper, these eigenvalues are all in the range $[0,1]$. The system is therefore

- asymptotically stable if

$$1-\alpha_1-\delta_1 < 1, \qquad 1-\alpha_2-\delta_2 < 1, \qquad 1-\beta_3-\gamma_3 < 1;$$

- stable (not asymptotically) if $\alpha_1 = \delta_1 = 0$ and/or $\alpha_2 = \delta_2 = 0$ and/or $\beta_3 = \gamma_3 = 0$.

There is no selection of the parameters which renders the system unstable. To see this, note that the $A$ matrix cannot have eigenvalues outside the unity disk, hence the system may be unstable only if there are eigenvalues at 1 (considering that all eigenvalues are real and positive) with geometric multiplicity larger than one. This requires at least two eigenvalues at 1 and a corresponding nonzero off-diagonal element to create an *unstable block*. Note, however, that if (for example) $1-\alpha_1-\delta_1 = 1$ and $1-\alpha_2-\delta_2 = 1$ then, necessarily, $\alpha_1 = \delta_1 = \alpha_2 = \delta_2 = 0$. As a result, all eigenvalues at 1 have geometric multiplicity one and the system cannot be unstable.

*While conditions for asymptotic stability have been correctly identified by the vast majority of the students, the discussion on stability/instability has been very imprecise/incomplete: for example almost all students have overlooked/neglected the role of geometric multiplicity.*

Answers

[ 6 marks ]

d) If $\alpha_1 = 0$ then the reachability matrix has rank equal to one. The system is already decomposed into reachable subsystem (with state $x_1$) and unreachable subsystem (with state $(x_2, x_3)$). As a result, all states described by the equation $\bar{x} = [\bar{x}_1,\ 0,\ 0]^T$, with $\bar{x}_1$ arbitrary, are reachable from the origin.

The system is controllable if and only if all the unreachable modes are at 0. The unreachable modes are the eigenvalues of the matrix

$$A_{nr} = \begin{bmatrix} 1-\alpha_2-\delta_2 & \beta_3 \\ \alpha_2 & 1-\beta_3-\gamma_3. \end{bmatrix}$$

These eigenvalues are both at zero provided $(1-\alpha_2-\delta_2)+(1-\beta_3-\gamma_3) = 0$ and $(1-\alpha_2-\delta_2)(1-\beta_3-\gamma_3)-\alpha_2\beta_3 = 0$. Nothing the constraints on the parameter the above equation is equivalent to

$$\alpha_2+\delta_2 = 1, \qquad \beta_3+\gamma_3 = 1, \qquad \alpha_2\beta_3 = 0 \tag{1.1}$$

In summary, if $\alpha_1 = 0$ the system is controllable if and only if the conditions (1.1) hold.

*Most students have failed to see that the system is already in* decomposed *form. They have therefore failed to identify the set of states that can be reached from zero and explicit conditions for controllability.*

[ 6 marks ]

Answers

2. a) The reachability matrix is

$$\mathscr{R} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix},$$

hence the system is reachable. Note that one could infer reachability of the system without computing the reachability matrix and simply observing that the matrix $A$ has a pair of complex conjugate eigenvalues and the rank of the reachability matrix is not smaller than one (since $B \neq 0$).

*All students have given the correct answer to this question.*

[ 2 marks ]

b) To begin with note that the characteristic polynomial of $A$ is $p(\lambda) = \lambda^2 + 1$, hence $a_1 = 0$ and $a_2 = 1$.

i) The desired characteristic polynomial is $p_d(\lambda) = (\lambda+1)^2 = \lambda^2 + 2\lambda + 1$, hence $a_{d1} = 2$ and $a_{d2} = 1$.

*Some students have failed to relate the position of the closed-loop eigenvalues with the desired closed-loop characteristic polynomial.*

[ 2 marks ]

ii) Note that (observe that $\Omega = I$)

$$K_{bg} = -\begin{bmatrix} 2 & 0 \end{bmatrix} I \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix} = -\begin{bmatrix} 1 & 1 \end{bmatrix}$$

and that

$$A + BK_{bg} = \begin{bmatrix} -1 & 0 \\ -2 & -1 \end{bmatrix},$$

which has the desired eigenvalues.

*Some, rare, mistakes in applying the given formula.*

[ 4 marks ]

iii) Note that

$$K_a = -\begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix} = -\begin{bmatrix} 1 & 1 \end{bmatrix}.$$

This is clearly the same state feedback gain matrix as the one in part b.ii) since the matrix $K$ assigning desired eigenvalues to a single-input controllable system is unique.

*Some students have obtained two different state feedback matrices in parts b.ii) and b.iii) and have not realized that, for the given system, the state feedback matrix assigning the eigenvalues to the desired location is unique.*

[ 4 marks ]

Answers

iv) If one commits the indicated error in the computation of the inverse of the reachability matrix one has, respectively,

$$\hat{K}_{bg} = -\begin{bmatrix} 1+2\varepsilon & 1 \end{bmatrix},$$

yielding the closed-loop characteristic polynomial

$$\lambda^2 + (2+2\varepsilon)\lambda + (1+2\varepsilon),$$

and

$$\hat{K}_{a} = -\begin{bmatrix} 1-2\varepsilon & 1 \end{bmatrix},$$

yielding the closed-loop characteristic polynomial

$$\lambda^2 + (2-2\varepsilon)\lambda + (1-2\varepsilon).$$

As a result, $\hat{K}_{bg}$ guarantees closed-loop stability for $\varepsilon > -1/2$, whereas $\hat{K}_{a}$ guarantees closed-loop stability for $\varepsilon < 1/2$. In this case the level of robustness against the indicated error in the computation of the inverse of the reachability matrix is essentially the same.

*Some, rare, mistakes in the computation of the perturbed closed-loop characteristic polynomials, but overall most students have provided correct robustness bounds.*

[ 8 marks ]

Answers

3. a) The reachability matrix of the system is

$$\mathscr{R} = \begin{bmatrix} 0 & 1 \\ 1 & a_{22} \end{bmatrix}$$

and the observability matrix is

$$\mathscr{O} = \begin{bmatrix} c_1 & c_2 \\ c_2 a_{21} & c_2 a_{22} + c_1 \end{bmatrix}.$$

Hence the system is reachable regardless of the value of the parameters (note that the system is in reachability canonical form) and observable provided

$$c_1^2 + c_1 c_2 a_{22} - c_2^2 a_{21} \neq 0.$$

*Incorrect entries, incorrect computation of the determinant/rank of $\mathscr{R}$ and/or $\mathscr{O}$, and/or incorrect conclusions.*

[ 2 marks ]

b) Note that

$$y(0) = Cx(0), \qquad y(1) = CAx(0) + CBu(0),$$
$$y(2) = CA^2x(0) + CABu(0) + CBu(1),$$
$$y(3) = CA^3x(0) + CA^2Bu(0) + CABu(1) + CBu(2),$$

yielding the equations

$$y(0) = c_2, \qquad y(1) = c_1 + c_2 + c_2 a_{22},$$
$$y(2) = c_2 a_{21} + (a_{22}c_2 + c_1)a_{22} + c_2 a_{22} + c_1 - c_2,$$
$$y(3) = c_2 a_{21} a_{22} + (a_{22}c_2 + c_1)(a_{22}^2 + a_{21}) + c_2 a_{21} + (a_{22}c_2 + c_1)a_{22} - c_2 a_{22} - c_1.$$

*This is a bookkeeping question in which some students have been inaccurate, hence have derived incorrect formulae.*

[ 6 marks ]

c) Solving the equations $y(0) = 1$, $y(1) = -1$, $y(2) = -3/2$, and $y(3) = 3$ as indicated in the exam paper one has

$$c_2 = 1, \qquad c_1 + a_{22} = -2, \qquad a_{21} - 2a_{22} = 3/2 \qquad a_{22} = -1,$$

hence $a_{21} = -1/2$, $a_{22} = -1$, $c_1 = -1$ and $c_2 = 1$. For these values of the parameters the system is observable.

*Incorrect formulae from part b) have propagated to this part and resulted in incorrect numerical values. Some students have also failed to see that the equations in part b) can be solved one by one, that is they form a* triangular *system of equations.*

[ 6 marks ]

Answers

d) Solving the same equation as in part c) with the new values of the $y$ sequence yields $a_{21} = a_{22} + 1$, $c_1 = -1 - a_{22}$ and $c_2 = 1$. Note that for these values of the parameters the system is not observable. One is therefore able to *identify* only the observable part of the system, whereas several unobservable parts, parameterized by $a_{22}$, are compatible with the observed output sequence.

*Incorrect formulae from part b) have also propagated to this part and resulted in incorrect numerical values. Several students have failed to see the relation between non-unicity of the solution and loss of observability.*

[ 6 marks ]

Answers

4. a) The equations of the interconnected system are $\dot{x} = Ax + B(K_{sf}x + K_I v)$, $\dot{v} = r - Cx$, and $y = Cx$, yielding the matrices

$$A_{int} = \begin{bmatrix} A + BK_{sf} & BK_I \\ -C & 0 \end{bmatrix}, \qquad B_{int} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad C_{int} = \begin{bmatrix} C & 0 \end{bmatrix}.$$

*Incorrect entries for the matrices of the closed-loop system.*

[ 2 marks ]

b) A direct computation yields

$$A_{aug} = \begin{bmatrix} A & 0 \\ -C & 0 \end{bmatrix}, \qquad B_{aug} = \begin{bmatrix} B \\ 0 \end{bmatrix},$$

*Incorrect entries for the matrices of the augmented system.*

[ 4 marks ]

c) i) The reachability pencil associated to the pair $\{A_{aug}, B_{aug}\}$ is

$$[sI - A_{aug} \quad B_{aug}] = \begin{bmatrix} sI - A & 0 & B \\ C & s & 0 \end{bmatrix}.$$

*Incorrect entries for the reachability pencil, in particular some students have failed to realize that the* $(2,2)$ *block is a of dimension* $1 \times 1$.

[ 2 marks ]

ii) Let $\bar{s}$ be a non-zero eigenvalue of the matrix $A$ and assume the pair $\{A, B\}$ is reachable. Then

$$\text{rank } [sI - A_{aug} \quad B_{aug}] = 1 + \text{rank } [sI - A \quad B] = n + 1,$$

where the first equality is the result of the fact that the second column of the reachability pencil has only one nonzero entry (the entry $\bar{s}$) and the second equality is obtained by reachability of the pair $\{A, B\}$.

*Incorrect computation of the rank, despite the fact that similar* objects *have been discussed in class.*

[ 4 marks ]

iii) The matrix $A_{aug}$ is lower triangular, hence its eigenvalues are the union of the eigenvalues of $A$ and 0, that is

$$\lambda(A_{aug}) = \lambda(A) \cap \{0\}.$$

Note that this implies that $A_{aug}$ has at least one eigenvalue at 0. It may have several eigenvalues at 0 if the matrix $A$ has eigenvalues at 0 as well.

*Some students have failed to realize that the matrix* $A_{aug}$ *is lower block triangular, hence the computation of the eigenvalue can be performed by inspection.*

[ 2 marks ]

Answers

iv) For zero eigenvalues of the matrix $A_{aug}$ one has

$$\text{rank}\,[-A_{aug} \quad B_{aug}] = \text{rank}\begin{bmatrix} -A & B \\ C & 0 \end{bmatrix},$$

where the equality is obtained nothing that for $s = 0$ the second column of the reachability pencil is identically zero. As a result, if the matrix on the right hand side has rank $n+1$ then the reachability pencil has rank $n+1$ for $s = 0$.

*This is a trivial question once the students realize that the given condition coincides with the requested condition for $s = 0$. Some have failed to* connect the dots.

[ 4 marks ]

v) Combining the results of parts c.iii) and c.iv), one concludes that the pair $\{A_{aug}, B_{aug}\}$ is reachable provided the original system is reachable and the matrix

$$\begin{bmatrix} -A & B \\ C & 0 \end{bmatrix}$$

has rank $n+1$. [ 2 marks ]

(The rank condition is equivalent to require that the transfer function $C(sI-A)^{-1}B$ does not have a zero at $s = 0$. It is not difficult to show that the indicated condition, that is reachability of the underlying system and the absence of a zero of the transfer function at $s =$ is necessary and sufficient for reachability of the pair $\{A_{aug}, B_{aug}\}$, although we have argued only the sufficiency.)

*Similar to the previous question, this one requires to perform a simple deduction. As a matter of fact the exercise is nothing else that a guided proof of a general statement. Some students have struggled to make a logic argument, even if similar arguments have been made in class.*

Answers